

IBM & Memorial University watsonx Hackathon

May 29th – 31st, 2026

Hackathon Guide

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IBM watsonx Hackathon

[AI agents](#) are intelligent systems that can perceive their environment, reason about it, and take actions to achieve specific goals. These agents can operate autonomously or collaboratively, making them ideal for solving complex real-world problems.

In this hackathon, you will design and build a proof-of-concept agentic AI solution using IBM watsonx products — a series of no-code and low-code products designed to orchestrate AI agents across business workflows. With watsonx, you can create, deploy, and manage intelligent agents and assistants that automate tasks, streamline processes, and enhance productivity.

This guide explains how to access your IBM Cloud environment, use watsonx, and build a proof-of-concept agentic solution for the hackathon.

The Hackathon Expectation

In this hackathon, you will design and build a proof-of-concept agentic AI solution using IBM watsonx, a low-code platform to build and deploy AI agents for the following theme:

AI Demystified — From idea to deployment

Every great idea deserves a faster path to reality. Use AI to simplify the steps that slow teams down — organizing work, connecting tools, or turning insights into action. Show how approachable, practical AI can make building and launching new solutions easier for everyone. Refer to example use cases to help you brainstorm ideas for building a solution.

A note on using other technologies

You may use any framework or technology to build your solution as long as you adhere to all product usage policies. However, to be eligible for judging, your solution must showcase IBM watsonx as a core component.

A note on data sets before you begin

Participants are required to bring their own datasets to build the solution aligning to your use case. As you collect data for your project, you'll want to use best practices. Here are helpful tips:

- Teams are responsible for ensuring data is compliant.
- Data from public websites may be used, if the terms allow for commercial use, but please keep a list of the websites you use.
- Do not use data or assets containing company confidential data, or any other data without permission from the data owner. Teams are responsible for getting approval.
- Do not use any client data.
- Do not use any data containing personal information (PI).

- Do not use data obtained from social media.

Sample Data sets you can use (optional)

- Newfoundland Geoscience - [Geoscience Resources of Newfoundland and Labrador](#)
- NL Mining Information [Geoscience - Energy and Mines](#)
- NL Oceans - [NL Oceans Data Hub – A hub for coastal and marine data and information in Newfoundland and Labrador](#)
- World Oceans dataset - <https://www.ncei.noaa.gov/products/world-ocean-database>
- Newfoundland and Labrador Oil and Gas Well inventory info [Well Data and Information | C-NLOPB Online Portal](#)
- General datasets <https://www.kaggle.com/datasets?fileType=csv>
- General Datasets: <https://huggingface.co/datasets>
- Newfoundland Government Open Data Sets <https://opendata.gov.nl.ca/>
- Stats Canada <https://www150.statcan.gc.ca/n1/en/type/data>

The Challenge

Create an AI-powered solution that transforms how organizations work, make decisions, or solve real-world problems in Newfoundland and Labrador. Your project must use at least one IBM watsonx service—and your imagination is the only limit.

Whether you're improving workflows, boosting productivity, or tackling a real industry challenge, your solution should demonstrate responsible, secure, and transparent AI.

What you're building:

Solve a business challenge in industries such as Ocean, Defense, Energy/Utilities, Mining, Manufacturing. Optional: Other sectors if justified.

Who you are:

Student at Memorial University looking to explore AI. No previous coding experience required!

Get started with IBM watsonx

To access and use IBM watsonx products for this hackathon, participants must create an IBMid to log into your IBM Cloud account. This account will provide the necessary environment to work with the supporting IBM products for this hackathon.

Creating an IBMid

You must have an IBMid to sign into IBM Cloud. With an IBMid, you can use your email login (use your school email) to log into all IBM products and services. If you don't have an existing IBMid, follow the procedure below to create one.

1. Create an IBMid by accessing the [Create your IBM account](#) page.
2. Enter all required information in the fields provided (using your school login).
3. Select **Next**. You will receive an email from IBM Security that contains a one-time verification code.
4. In the **verification token** field, enter the code that is provided in the email.
5. Click **submit**.
6. An email will be sent to you from IBM Security indicating that your IBMid account creation was successful and that your account is activated.

Accessing and utilizing IBM watsonx products

Once you and your team are assigned Cloud environments, be sure to accept the environment cluster invitation that will be sent to your email. It should look something like this:



Action required: You are invited to join an account in IBM Cloud

Source ID: 1779984915270

Type: Account

Update Time: 28 May 2026, 4:15 PM UTC

View notification (in subscribed accounts):

Hi Sofia.Jia@ibm.com,

IBM IBM Technology Zone Adhoc 01 invited you to join the following IBM Cloud account: itz-watsonx-event-002 (Account ID: 7d4b275ed9a2467091869b482a8f99f0).

[Join now.](#)

The link to join the account expires in 30 days.

Thank you,
IBM Cloud

In case of any issues, please contact [IBM Cloud support](#).

IMPORTANT: You **must** click the “Join now” to get access, and ensure you log into IBM Cloud using the same email you used for the IBMid setup. Not clicking the join now button or using a different email will not give you proper access to the environments!

To begin building your solution, explore the capabilities and resources for each IBM watsonx product enabled for this hackathon.

1. IBM watsonx Orchestrate

IBM watsonx Orchestrate is an intuitive, AI-powered platform that you can use to create, configure, and deploy intelligent agents that can automate business tasks. Whether you’re automating repetitive workflows or building complex multi-agent systems, the platform is designed to support users of all skill levels.

2. IBM watsonx.ai

A powerful AI studio that enables you to experiment with foundation models like IBM Granite through the Prompt Lab. It can be used to enhance your agents with natural language understanding, intelligent decision-making, and dynamic response generation — serving as an inference engine to make your AI agents smarter and more interactive.

1. IBM watsonx Orchestrate

After successfully joining the IBM Cloud account, you can access watsonx Orchestrate to begin working on the platform and building your solution.

With watsonx Orchestrate, you can:

- Build and deploy agents
- Automate tasks such as scheduling, data entry, and approvals
- Start small and scale up with advanced tools and integrations
- Explore the clean, guided UI that simplifies agent creation and management.

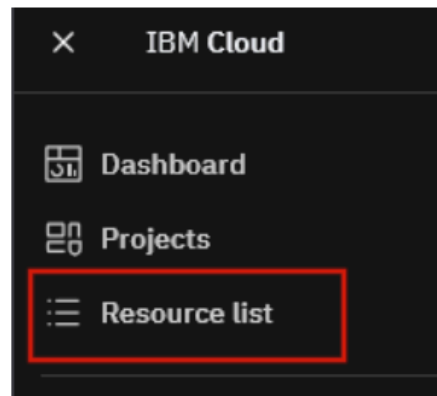
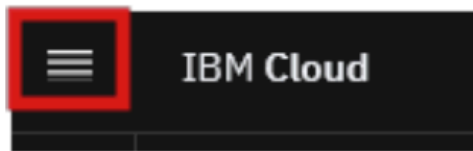
Explore a few demos on how to use watsonx Orchestrate.

These watsonx Orchestrate features/capabilities are out of scope for this hackathon:

- Build with AI (Preview)

Accessing watsonx Orchestrate

1. In your IBM Cloud account dashboard, select the Navigation menu on the top left of the dashboard and select the Resource list option.



2. Expand the AI / Machine Learning section and select watsonx Orchestrate service.
- 3.

^ AI / Machine Learning (7)				
	Watson OpenScale-itx	watsonx	Dallas (us-south)	watsonx.governance
	Watson Orchestrate-itx	itz-wxo-6a144957a88d04ff33d340	Dallas (us-south)	watsonx Orchestrate
	Watson Orchestrate-itx	itz-wxo-6a16be4b3edca554f672f1	Dallas (us-south)	watsonx Orchestrate

4. You will be navigated to watsonx-Hackathon Orchestrate service instance dashboard.
Select the Launch watsonx Orchestrate button.

[Resource list](#) /

Watson Orchestrate-itx [Add tags](#)

Manage

Service credentials

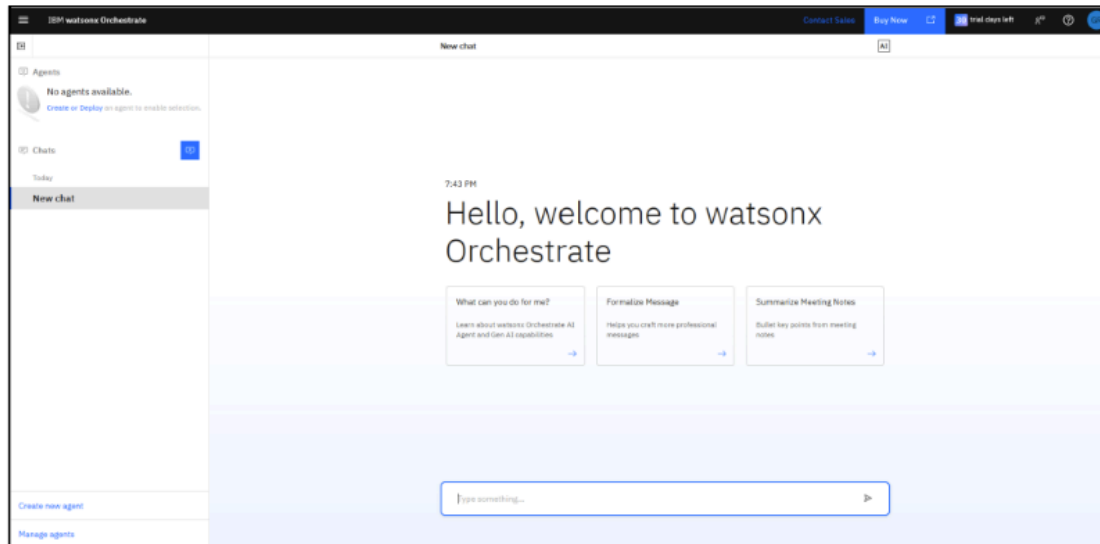
Plan

Start by launching the tool

[Launch watsonx Orchestrate](#)

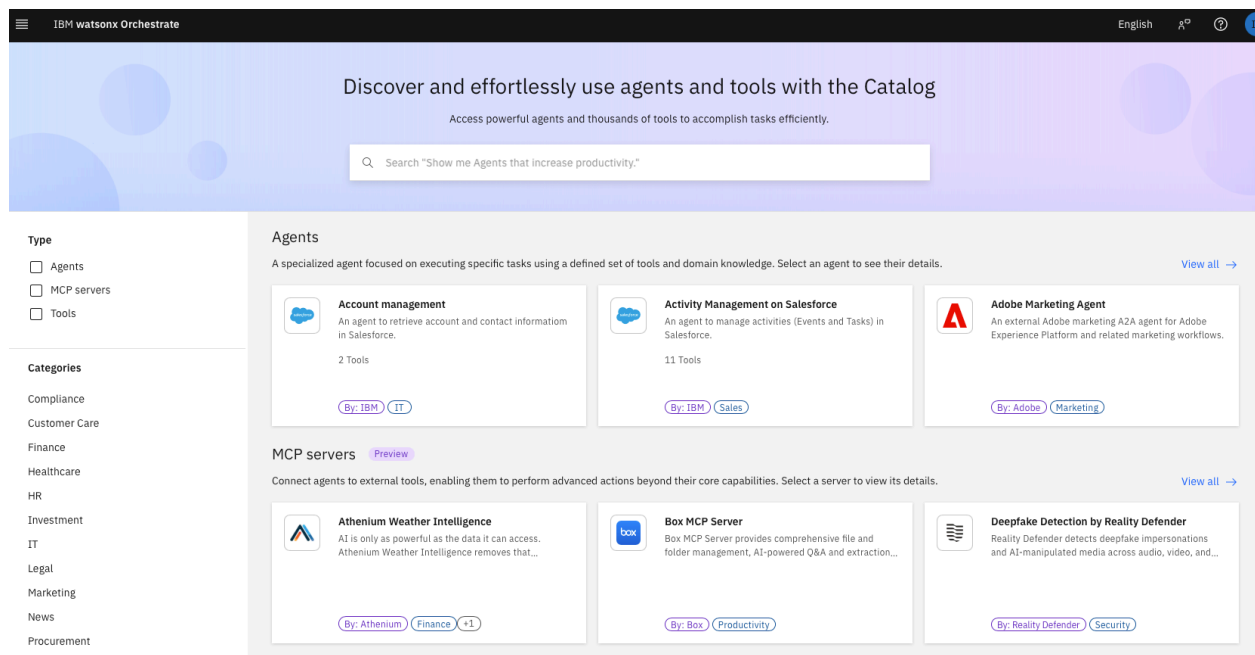
[Getting started tutorial](#)

5. You will be navigated to the watsonx Orchestrate platform with a welcome message and a new chat window.



Discover the catalog

The IBM watsonx Orchestrator catalog is your gateway to a rich collection of prebuilt AI agents and tools, designed to support a wide range of business functions and use cases. Whether you're looking to automate tasks, enhance productivity, or integrate with backend systems, the catalog helps you find the right solutions quickly and efficiently. Learn more about [discovering the catalog](#).



Building agents

In IBM watsonx Orchestrate, agents are a key component of the agentic AI framework, enabling you to create complex, dynamic systems that can adapt and respond to changing conditions.

By building agents, you can:

- Automate repetitive tasks
- Improve decision-making processes
- Enhance customer and employee experiences
- Increase operational efficiency

1. Prepare to build AI agents

Developing intelligent, scalable, and reliable agents in watsonx Orchestrate requires a strategic, multi-phase approach. It goes beyond writing prompts or connecting APIs. Successful agents need thoughtful planning, structured development, testing, and ongoing governance. The following overview breaks down each phase of agent development, highlighting key considerations. Learn more about [preparing to build AI agents](#).

2. Creating and customizing agents

Creating and customizing an AI agent involves defining its purpose and personality through thoughtful descriptions and selecting the most suitable foundation model that aligns with your brand or use case. You can also configure key elements like the welcome message and starter prompts to ensure the agent engages users effectively from the first interaction. Together, the configured components shape how the agent communicates, responds, and delivers value across conversations.

Learn how to create an [agent from scratch or from a template](#).

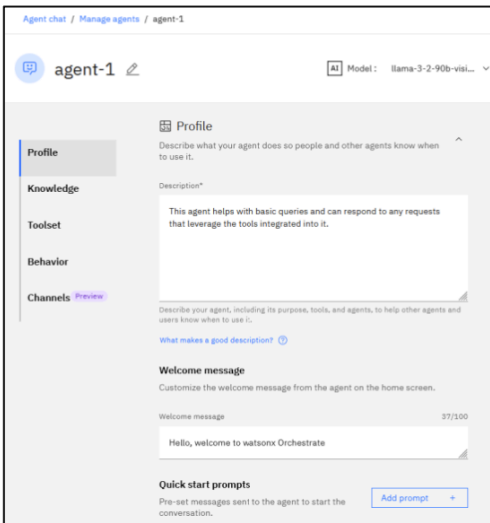
The screenshot shows a 'Create an agent' dialog box with two tabs: 'Create from scratch' (selected) and 'Create from template'. The 'Create from scratch' tab contains a text input field for 'Name' with the placeholder 'Name your agent', a text area for 'Description' with an example text 'Example: This agent helps answer customer questions about return and cancellation policies.', and a link 'What makes a good description?' with a help icon. At the bottom are 'Cancel' and 'Create' buttons.

Expanding your agent's capabilities

To enhance your agent's abilities, explore the following configuration options:

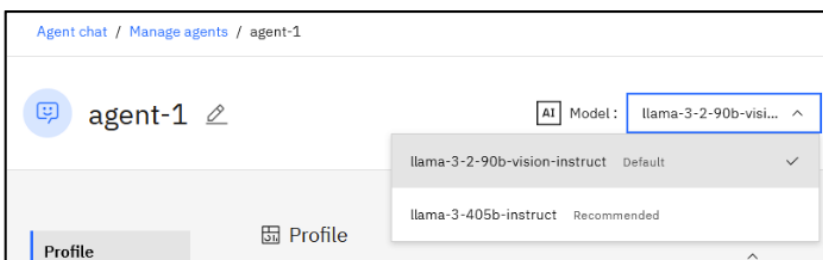
- **Define a profile**

Provide a clear and specific description of the agent's purpose. This helps in multi-agent orchestration by enabling accurate selection based on capabilities. See [Defining the description of your agent](#).



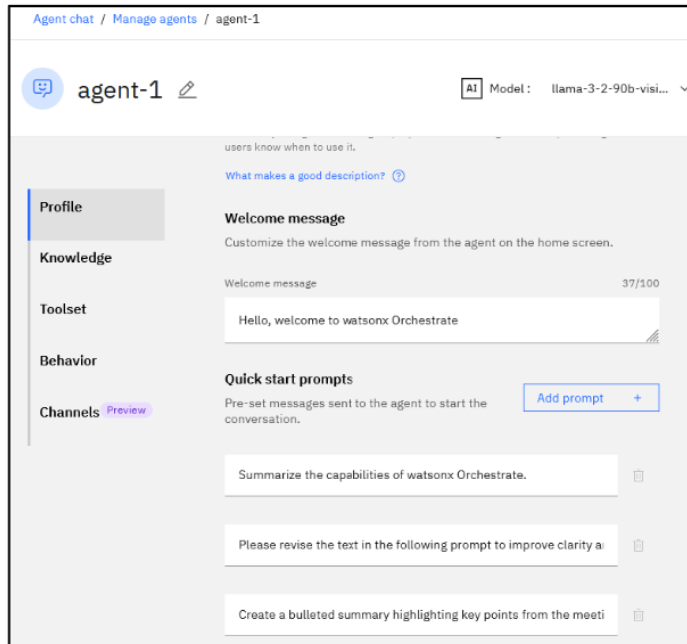
- **Choose an AI model**

Choosing the right AI model helps your agent understand user intent, reason through tasks, and generate reliable responses. Your choice directly affects accuracy, performance, and cost. See [Choosing an AI model](#).



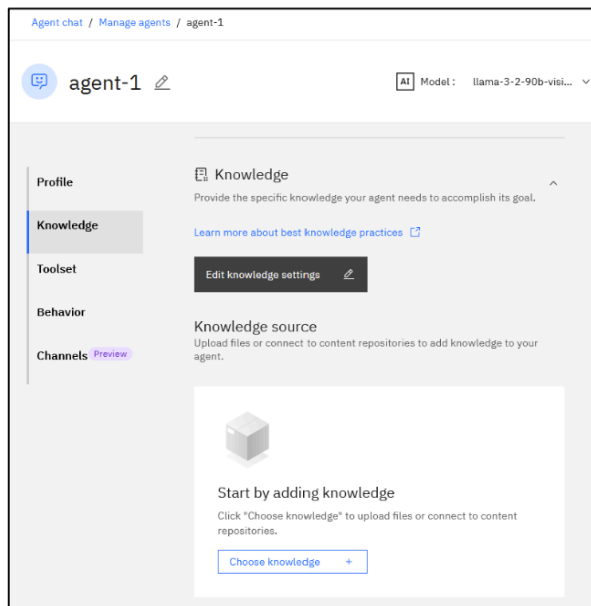
- **Customize the welcome message and starter prompts**

Customize the welcome message and starter prompts to guide users when they begin interacting with the agent. See [Customizing the welcome message and starter prompts](#).



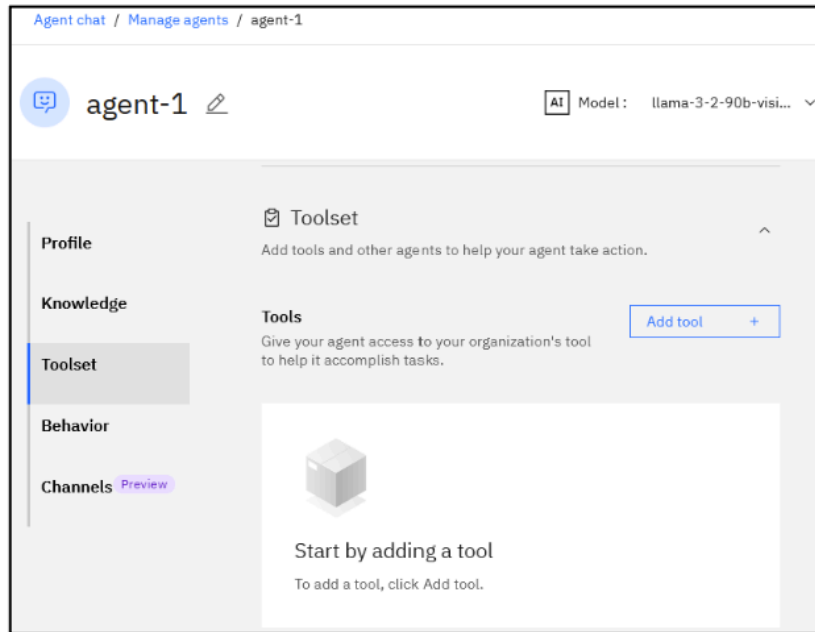
- **Add knowledge**

Enhance the agent's domain expertise by adding contextual knowledge from files or content repositories. See [Adding knowledge to agents](#).



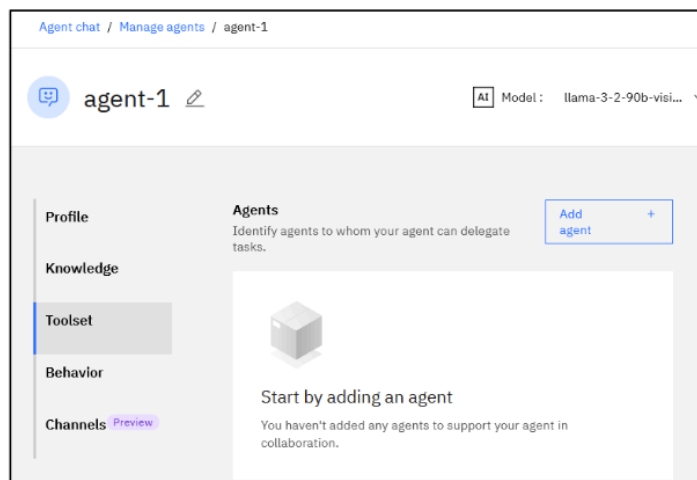
- **Add and manage tools**

Integrate tools to enable the agent to perform automated tasks such as retrieving data or sending emails. See [Adding and managing tools for agents](#).



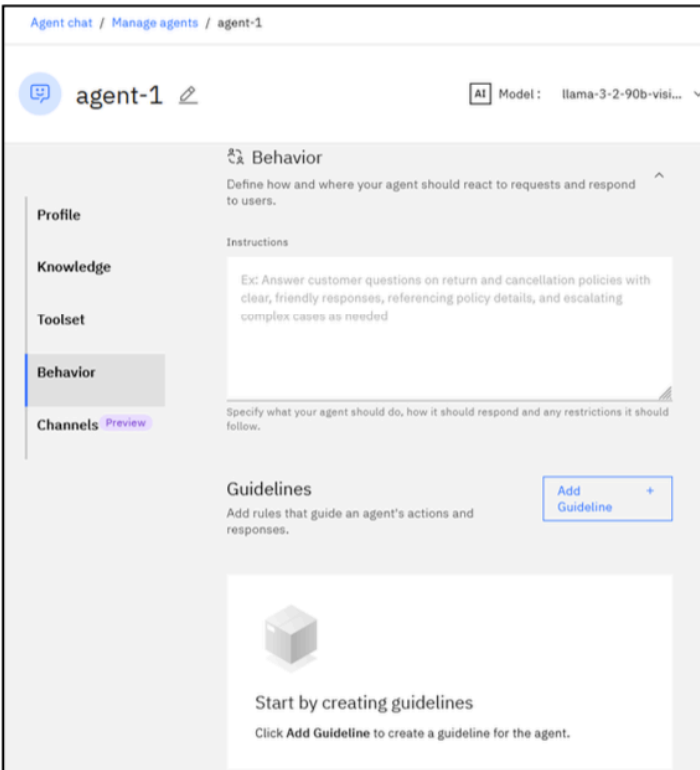
- **Orchestrate agents**

Enable multi-agent orchestration by adding collaborator agents that work together to achieve shared goals. See [Orchestrating agents](#).



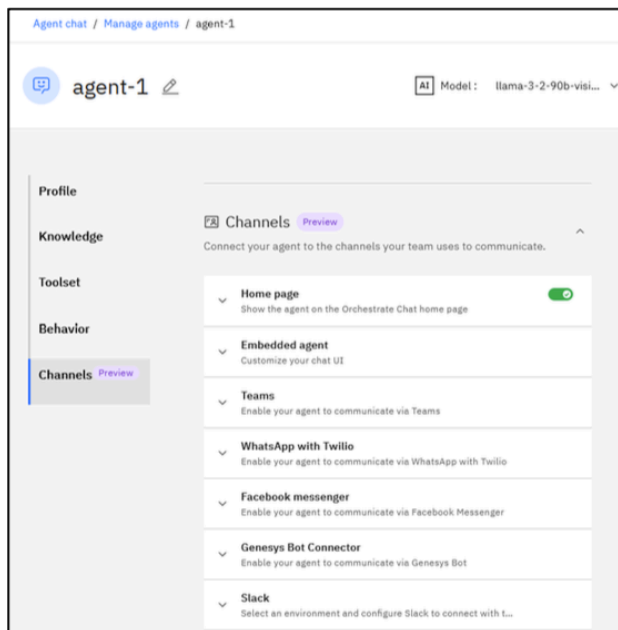
- **Add instructions**

AI agents operate based on a set of behavioral instructions that define how they respond to user inputs, perform tasks, and interact across different channels. These instructions serve as the agent's internal guidelines, helping it act consistently and effectively in various scenarios. See [Adding instructions to agents](#).



- **Connect agents to channels**

Channels are the interfaces through which you can interact with a conversational agent, such as messaging platforms and voice assistants. Integrate your agent with these channels to access services and support directly within your preferred environments to enable seamless and scalable communication. See [Connecting agents to channels](#).

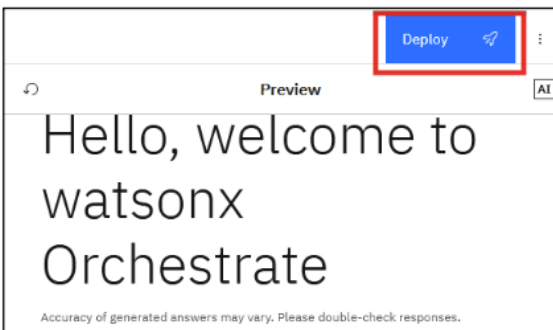


- **Configure rich responses**

Enable rich responses to incorporate multimedia and structured elements, making AI interactions clearer, more efficient, and engaging for users. See [Configuring rich responses from the AI assistant builder](#).

- **Deploy agents**

Finalize the setup by deploying the agent to make it available in live environments such as chat interfaces. See [Deploying agents](#).



- **Building agents using the ADK**

You can build powerful, customizable agents using the IBM watsonx Orchestrate Agent Development Kit (ADK). [Learn more about using the ADK](#).

3. Building tools

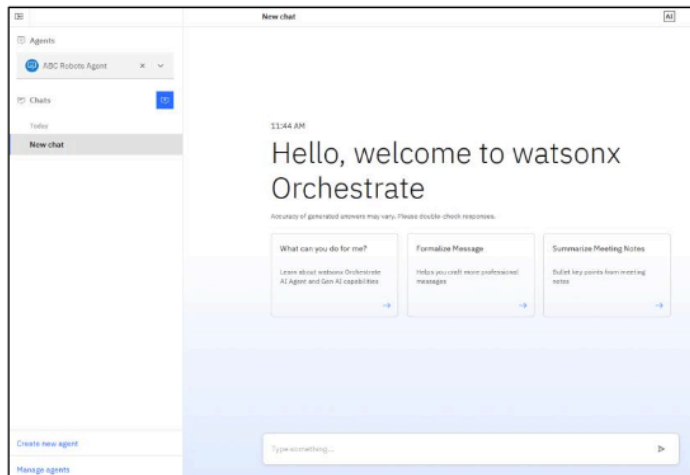
Tools act as interfaces to external capabilities, enabling agents to interact with systems, retrieve data, run calculations, or trigger workflows that go beyond their built-in reasoning. You can build new tools and edit existing tools for your agents to use.

Following are the different ways to add a tool:

- [Creating agentic workflows](#)
- [Adding tools to an agent](#)
 - Catalog
 - Local instance
- [Importing external tools](#)
 - MCP server
 - OpenAPI

Using agents in the chat

In IBM watsonx Orchestrate, agents collaborate to automate tasks and manage workflows. [Learn more using agents in Orchestrate Chat](#).



Managing app connections

To use the external applications within IBM watsonx Orchestrator, you must establish a connection between them which acts as bridge enabling communication between watsonx Orchestrator and the external applications. [Learn more about managing app connections and credentials.](#)

watsonx Orchestrator API

IBM watsonx Orchestrator provides a set of APIs to boost your experience using the features from the product.

- [Getting started with the API](#) — Begin using the API with practical, step-by-step guidance.
- [API reference](#) — Get the methods that you can use to call custom projects, or other resources from the AI chat.

Building AI assistants

In IBM watsonx Orchestrator, you build the AI assistant by using AI assistant builder. AI assistant builder is a chat interface builder that helps to deploy an engaging and embedded chatbot experience. AI assistant builder integrates the power of large language models (LLMs) and conversational capabilities of watsonx Assistant to enable responsive and interactive conversation between the users and watsonx Orchestrator.

To learn more, see [Building AI assistants in AI assistant builder.](#)

Quick start hands-on exercises

Try the quick start hands-on exercises for sample use cases to get started with using watsonx Orchestrate:

Important notes:

- You must use the hackathon provisioned IBM cloud account to access and use watsonx Orchestrate platform to try the below sample exercises.

watsonx Orchestrate SaaS sample exercises

- [Develop agents with no code using watsonx Orchestrate](#) — A hands-on guide for developing a multi-agent system in watsonx Orchestrate with no code that can communicate with the agents on various channels.
- [Creating intelligent, reusable agentic workflows on watsonx Orchestrate with no code](#) — A hands-on guide for creating AI-driven, human-in-the-loop workflows without writing code.
- [Build context-aware AI agents with watsonx Orchestrate and Astra DB](#) — AI agents are only as effective as their data, making precise retrieval essential for reliable results.
- [AgentOps in watsonx Orchestrate: Observability for Agents with Langfuse and IBM Telemetry](#) — A developer's guide for instrumenting and observing Agents with Langfuse and IBM Telemetry in watsonx Orchestrate.
- [Building and deploying prebuilt domain agents in watsonx Orchestrate](#) — Explore how to work with the prebuilt domain agents and integrate them into existing workflows.
- [Build a scheduled agentic workflow for daily notifications](#) — A practical example of building, importing, and scheduling an agentic workflow that automatically sends daily email notifications using watsonx Orchestrate.

- [**Multi-agent orchestration with watsonx Orchestrate**](#) — A hands-on journey to design, extend, and integrate AI agents by using Langflow, IBM Granite models, Astra DB, the AI Gateway, and a custom user interface on watsonx Orchestrate.

watsonx Orchestrate Agent Development Kit (ADK) sample exercises

- [**Getting Started with watsonx Orchestrate Agent Development Kit**](#) — IBM watsonx Orchestrate includes the Agent Development Kit (ADK) which is a set of developer-focused tools for building, testing, and managing agents. With the ADK, developers get the freedom and control to design powerful agents using a lightweight framework and a simple CLI. You can define agents in clear YAML or JSON files, create custom Python tools, and manage the entire agent lifecycle with just a few commands.
- [**Build an AI agent with Langflow, Granite 4.0 models, and watsonx Orchestrate**](#) — A hands-on guide for creating Langflow tools, consuming them with watsonx Orchestrate Agent Development Kit, and using IBM Granite 4.0 Micro model for inference.
- [**Connecting to MCP tools with watsonx Orchestrate**](#) — A hands-on guide for creating MCP tools and consuming them with watsonx Orchestrate Agent Development Kit.
- [**Extend Your AI Agents with External LLMs Using watsonx Orchestrate and AI Gateway**](#) — A hands-on guide to bring your own model (BYOM) into watsonx Orchestrate using AI Gateway.
- [**Create consistent evaluation workflows for AI agents**](#) — A hands-on guide to setup and use a structured evaluation strategy to test, benchmark, and improve AI agent performance.

2. IBM watsonx.ai

After successfully joining the IBM Cloud account, you can access watsonx.ai to begin working on the platform and building your solution.

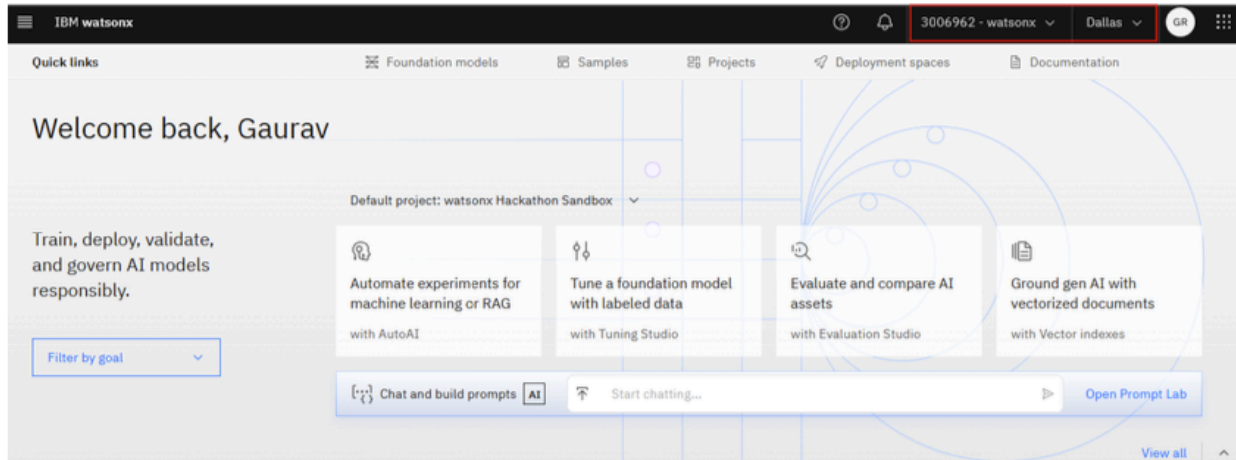
Note on IBM watsonx.ai service usage

The hackathon provisioned IBM Cloud account will be deactivated after the completion of the hackathon. Please plan to save your work at the end of the hackathon.

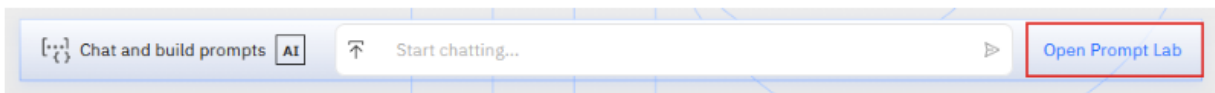
Access Prompt Lab on watsonx.ai

After successfully joining the IBM Cloud account, you can now access the Prompt Lab on watsonx.ai platform to work with the AI models supported on the platform and build your solution.

1. Log in to the [watsonx.ai platform](#) with the email you used to access your IBM Cloud account.
2. After successful authentication, you will see "Welcome to watsonx" widget. You can either take the tour or skip it.
3. Next, you will see the watsonx.ai dashboard. Ensure the name of the account is "xxxxxxx – watsonx".

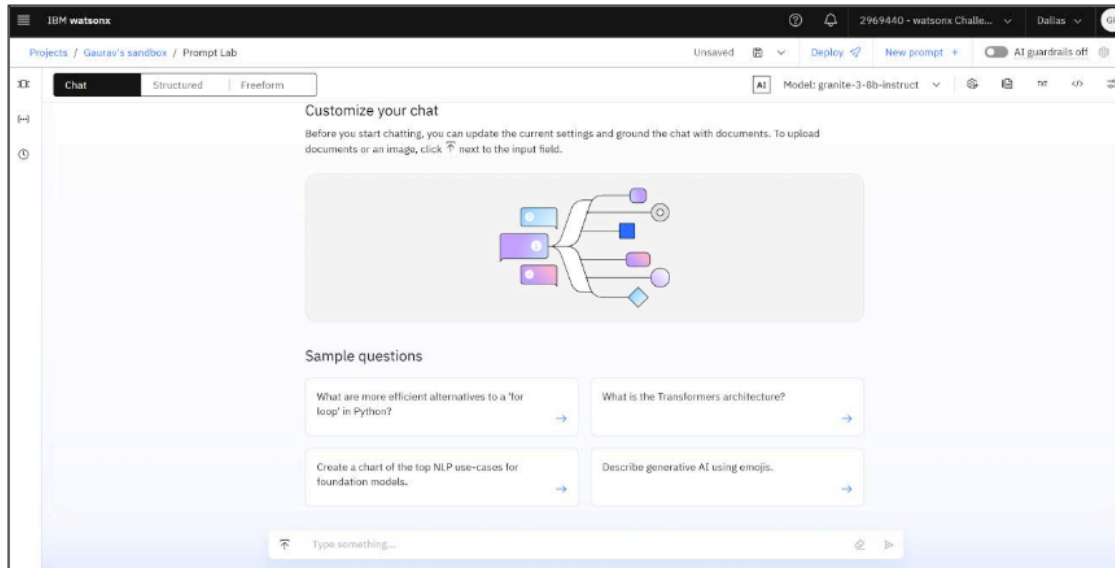


4. Select **the Open Prompt Lab** button on the "Chat and build prompts with foundational models" widget.



5. The "Welcome to Prompt Lab" tour will be displayed. You can take the tour to get a quick introduction or skip it.

6. The Prompt Lab Editor opens with a chat window to get you started with the prompt session.



Work with the watsonx.ai Prompt Lab

The watsonx.ai Prompt Lab is an easy-to-use prompt engineering interface where you can experiment with prompting different AI foundation models, explore sample prompts, tune model parameters, integrate applications with an API endpoint, and save and share your best prompts.

[Take a tour of the Prompt Lab](#) and try the [interactive demo](#).

You can access and use the AI models to build your innovative solution using Prompt Lab.

Prompt Lab editor

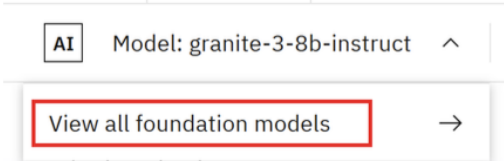
In the Prompt Lab, you can experiment with prompting different foundation models, explore sample prompts, as well as save and share your best prompts. The Prompt Lab editor is a great place to experiment and iterate with your prompts. [Try the quick start lab](#).

However, you can also prompt foundation models in watsonx.ai programmatically. Refer to "Programmatic access (API/SDK)" section.

Selecting an AI model

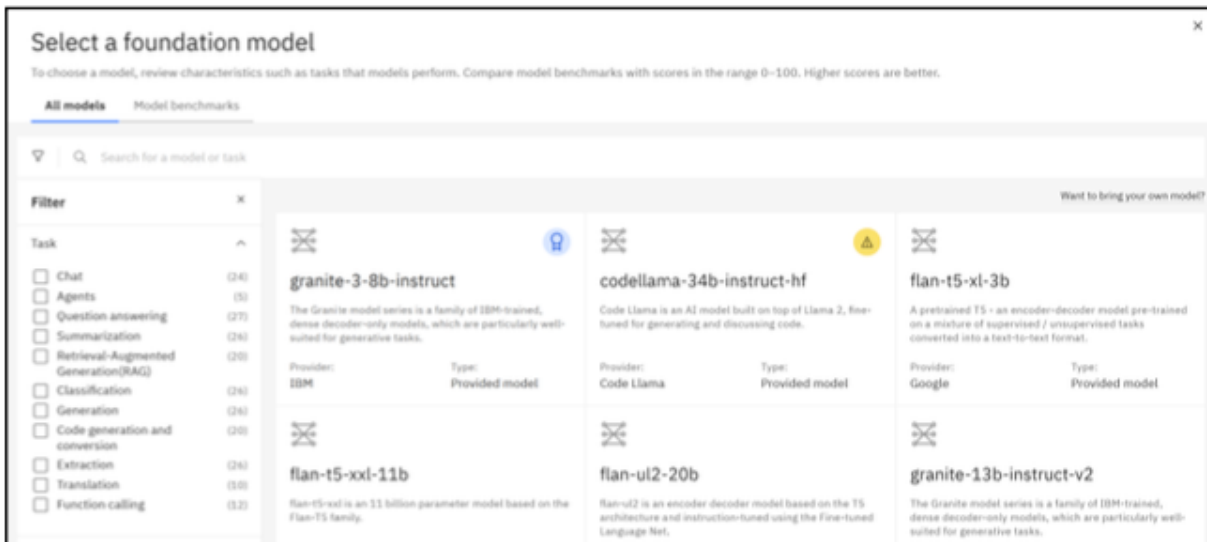
A default AI model will be pre-selected in the Prompt Lab editor. You can either use the same model or change to a different model. To select a different model:

1. Select the AI Model drop-down menu at the top-right of the editor and select View all foundation models.



2. A foundation model selection widget appears. Clear the filters to see all the available models. You can use the filters to choose the right model for your solution building. You can select a model tile to learn about the model and use it. If you are limited to only "Chat" supported models, change the Prompt Lab editor to Structured or Freeform view and try selecting the models to see all the available model options.

To understand how models can address your use case, including information on model modalities, supported languages, tuning, and indemnification, see our product documentation on [choosing a model](#).



Note: Bigger models are not always better. [Learn](#) why smaller models can be better and more cost effective.

Programmatic access (API/SDK)

You can inference the watsonx.ai models with API or SDK requests.

Developer access information

To use the supported watsonx.ai APIs/SDKs, you will need three values: a **project ID**, an **endpoint URL** and an **API key**.

- Go to [watsonx.ai home page](#).
- Scroll down to the "Developer access" section.

Developer access ⓘ

Project or deployment space Project ID

Project or deployment space 00000000-0000-0000-0000-00000000 ⓘ

watsonx.ai URL

<https://us-south.ml.cloud.ibm.com> ⓘ

Used to call watsonx.ai APIs such as LLM inferencing, embedding, training, and chatting.

[Create API key](#) + [Manage IBM Cloud API keys](#) →

- Select the Project or deployment space drop-down and choose the watsonx Hackathon Sandbox option if you are working with that project, or select any other listed project that you are working with. A project ID will be displayed.

Project or deployment space Project ID

watsonx Hackathon Sandbox 9f5501-0000-0000-0000-00000000 ⓘ

- A default watsonx.ai endpoint URL will be displayed for the Dallas region. Ensure the region is always set to Dallas at the top right of the watsonx.ai home page.
- Select the Create API key button. A Create API key widget will be displayed. Enter a name, provide optional description and choose the "Disable the leaked key" option. Click the Create button.

Create API key ×

You need an IBM Cloud API key to create the IAM bearer token for authenticating your API requests. [Learn more](#)

Name

watsonx API key

Description (optional)

Enter name

Choose what to do if this key is leaked:

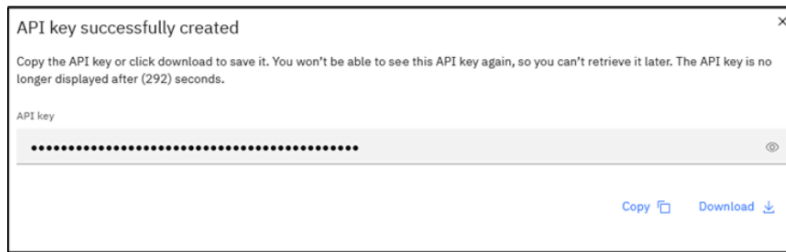
☒ Disable the leaked key

☐ Delete the leaked key

☐ Nothing

[Cancel](#) [Create](#)

- An API key will be created successfully. Copy the API key and save it safely to use for calling the API/SDK. You can also download and save the file in a secure path in your system.



watsonx.ai programmatic options

There are multiple options to help you get started using watsonx.ai APIs/SDKs.

Option 1: Prompt Code on Prompt Lab

Refer to the access prompt code instructions to learn how to quickly get access to the text generation API within the watsonx.ai Prompt Lab.

Option 2: Different watsonx.ai API capabilities

Explore and leverage different watsonx.ai API capabilities in your solution.

- [Chat](#)
- [Agent-driven chat](#)
- [Tool calling](#)
- [Text generation](#)
- [Time series](#)
- [Text rerank](#)
- [Embeddings](#)
- [Text extraction](#)

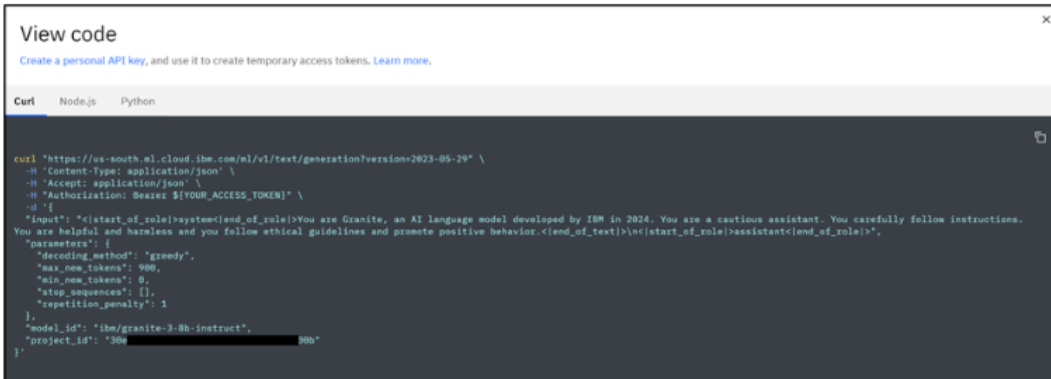
Refer to supported API functionality by AI model in [the watsonx.ai documentation](#).

Access the prompt code (API) from Prompt Lab editor

To prompt an AI model programmatically, you can view and copy the prompt code by selecting the **View code** icon `</>` at the top-right of the prompt lab editor.



The prompt code is available as a Curl, Node.js and Python.

A screenshot of a 'View code' window. At the top, it says 'View code' and 'Create a personal API key, and use it to create temporary access tokens. Learn more.' Below this is a tab bar with 'Curl', 'Node.js', and 'Python'. The 'Curl' tab is selected, showing a cURL command for the IBM Granite API. The command is a POST request to 'https://us-south.ml.cloud.ibm.com/ml/v1/text/generation/version=2023-05-29'. It includes headers for 'Content-Type: application/json', 'Accept: application/json', and 'Authorization: Bearer \${YOUR_ACCESS_TOKEN}'. The body is a JSON object with 'input' (a system and assistant role prompt), 'parameters' (decoding_method: greedy, max_new_tokens: 900, min_new_tokens: 0, stop_sequences: [], repetition_penalty: 1), 'model_id' (ibm/granite-3-8b-instruct), and 'project_id' (30a).

```
curl "https://us-south.ml.cloud.ibm.com/ml/v1/text/generation/version=2023-05-29" \
-H 'Content-Type: application/json' \
-H 'Accept: application/json' \
-H 'Authorization: Bearer ${YOUR_ACCESS_TOKEN}' \
-d '{
  "input": "(start_of_role)system(end_of_role)You are Granite, an AI language model developed by IBM in 2024. You are a cautious assistant. You carefully follow instructions. You are helpful and harmless and you follow ethical guidelines and promote positive behavior.(end_of_text)(start_of_role)assistant(end_of_role)",
  "parameters": {
    "decoding_method": "greedy",
    "max_new_tokens": 900,
    "min_new_tokens": 0,
    "stop_sequences": [],
    "repetition_penalty": 1
  },
  "model_id": "ibm/granite-3-8b-instruct",
  "project_id": "30a"
}'
```

You will require an IAM access token to authorize the prompt code and need to replace **`\${YOUR\_ACCESS\_TOKEN}`** placeholder in the prompt code. You can create an IAM access token using an API key.

- **API key:** Refer to Developer access information to get an API key.
- **Generate IAM Access Token:** Programmatically generate an IAM access token with the API key using the following cURL command:

```
curl -X POST 'https://iam.cloud.ibm.com/identity/token' -H 'Content-Type: application/x-www-form-urlencoded' -d 'grant_type=urn:ibm:params:oauth:grant-type:apikey&apikey=MY_APIKEY'
```

- **curl -X POST** → Specifies an HTTP POST request.
- **URL ("https://iam.cloud.ibm.com/identity/token")** → The endpoint to request an authentication token from IBM Cloud.
- **-H "Content-Type: application/x-www-form-urlencoded"** → Sets the request header to indicate that the data is sent in form-encoded format.
- **-d (Data Payload)** → Sends the required data:
 - **grant_type=urn:ibm:params:oauth:grant-type:apikey** → Specifies the OAuth grant type as API Key.
 - **apikey=MY_IBM_CLOUD_API_KEY** → Replace MY_IBM_CLOUD_API_KEY with your actual IBM Cloud API key.

Expected Response:

```
{
  "access_token": "eyJhbGciOiJIUz....sgrKIi8hdFs",
  "refresh_token": "not_supported",
  "token_type": "Bearer",
  "expires_in": 3600,
  "expiration": 1473188353,
  "scope": "ibm openid"
}
```

Note: An IAM token is valid for up to 60 minutes, and it is subject to change. When a token expires, you must generate a new one. Use the property "expires_in" for the expiration of the IAM token that you have just created.

watsonx.ai AI agent libraries and tutorials

Explore the watsonx.ai supported AI agent framework libraries and tutorials to help you get started building your AI agent solution.

- [LangChain](#)
- [LangGraph](#)
- [LlamaIndex](#)
- [CrewAI](#)
- [BeeAI](#)
- [AutoGen](#)
- [Python SDK](#)
- [Node.js SDK](#)

Quick start hands-on exercises

Try the quick start exercises and notebooks for sample use cases to get started with using watsonx.ai.

Important notes:

- Refer to developer access information section to use watsonx.ai credentials as you try the exercises.
- Some of the exercises could include the usage of old model version. You can replace them with newer versions for better performance and output. To check the latest supported AI models on watsonx.ai, either follow selecting an AI model on Prompt Lab or refer to supported foundation models on watsonx.ai.
- The hackathon provisioned cloud accounts do not support solution deployment. You can run your solution deployment locally on your machine and showcase them in your submissions.
- If you are [using Jupyter Notebook editor on watsonx.ai](#), consider selecting a lower runtime environment to avoid high resource consumption and quickly depleting your credits. Notebook runtimes are billed based on Capacity Unit Hours (CUH) at a rate of \$1.02 USD per CUH. Learn more about [capacity unit hours and watsonx.ai Studio pricing plans](#).

watsonx.ai Prompt Lab app templates

- [LangGraph LLM app template with function calling capabilities](#) (base template)
- [LlamaIndex Workflow LLM app template with function calling capabilities](#) (base template)
- [CrewAI LLM app template with function calling capabilities](#) (base template)
- [arXiv Research agent](#) (community template)
- [Agentic RAG LangGraph template](#) (community template)

BeeAI Agent Framework

- [BeeAI framework examples](#)

LangChain and LangGraph

- [Create a LangChain AI Agent in Python using watsonx](#)
- [Build a RAG agent using LangGraph to answer complex questions](#)
- [Build a LangChain agentic RAG system using the Granite model in watsonx.ai](#)
- [Use watsonx, and LangChain Agents to perform sequence of actions](#)
- [Use watsonx, and LangChain to make a series of calls to a language model](#)
- [arXiv Research agent](#)
- [Base LangGraph LLM app template with function calling capabilities](#)

LlamaIndex

- [Use watsonx and LlamaIndex for Text-to-SQL task](#)
- [Use watsonx, and `llama-3-1-70b-instruct` and LlamaIndex to make simple chat conversation and tool calls](#)
- [LlamaIndex Workflow LLM app template with function calling capabilities](#)

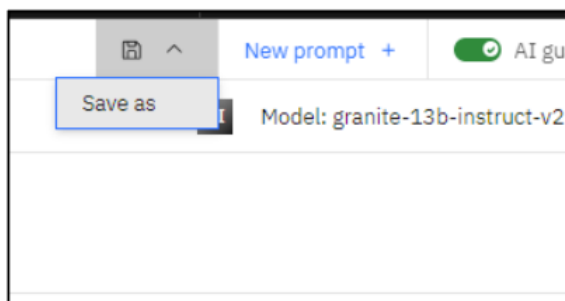
CrewAI

- [Leveraging CrewAI and IBM watsonx](#)
- [Build an agentic framework with CrewAI memory, i18n, and IBM watsonx.ai](#)
- [Base CrewAI LLM app template with function calling capabilities](#)

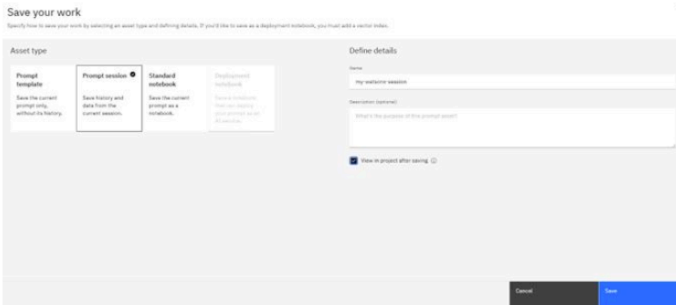
Save your Prompt Lab session

You can save your Prompt Lab editor session for later use.

1. At the top of the Prompt Lab screen, select the Save work dropdown button and then select the Save as option.



2. A Save your work widget will appear. Select Prompt session under the Asset type option.
3. Enter a name and check the View in project after saving option under the Define details section.
4. Finally, click the Save button. Once you save, you will see the saved work under the Assets tab.



You can also save your work as:

- **Prompt template** to save only the current prompt without its history and selecting a Task suitable for your prompting.
- **Notebook** to continue prompting on a Jupyter Notebook environment. Prior knowledge of notebooks and Python programming language would be helpful to work with a Jupyter notebook. [Read more about notebooks.](#)

Save your work on watsonx.ai

Make sure to save any work you want to retain for your records. IBM Cloud accounts will be deactivated at the end of the hackathon. Follow the steps below to save your work:

5. Go to your project's "Overview" tab.
6. Select the "Export or import project" drop down below the Bell icon in the top menu bar.
7. Click the "Export project" option. This will open "Export project to desktop" screen.
8. Select all the assets shown in your project (Work saved as Project session cannot be exported) and click "Export" on the bottom-right of the screen.
9. The next screen will ask for confirmation that all sensitive information has been removed.
10. Click on "Continue export".
11. The download (zip) will be initiated, and the file will be saved on your computer.

Appendix: Example use cases

You are not limited to these ideas, but here are several examples for how you could apply IBM watsonx Orchestrate to solve a specific issue for this hackathon:

- **Port Operations Autopilot:** Create an agent that helps a port run more smoothly by coordinating ship arrivals, cargo unloading, and staff schedules. The agent could read ship arrival data, predict delays, notify port workers, and update scheduling tools automatically. It might also generate daily summaries showing which ships arrived, which are delayed, and what tasks need attention. This reduces manual coordination and helps students understand how automation improves logistics.
- **Ocean Research Data Organizer:** Build an agent that collects data from ocean sensors (temperature, salinity, pollution levels), organizes it into charts, and drafts summaries for student research projects. It could also notify teams when unusual readings appear, helping students see how AI supports environmental science.
- **Equipment Health & Maintenance:** Develop an agent that monitors equipment logs (vehicles, radios, drones), predicts when maintenance is needed, and automatically creates service tickets. It can also notify supervisors when equipment is unavailable. This helps students understand predictive maintenance and operational reliability.
- **Smart Outage Detection & Response Agent:** Create an agent that monitors sensor data from the electrical grid and identifies early signs of outages. It could alert technicians, create work orders, and update outage maps automatically. Students learn how AI improves public infrastructure reliability.
- **Renewable Energy Performance Monitor:** Develop an agent that analyzes data from solar panels or wind turbines, identifies underperforming equipment, and drafts maintenance recommendations. It could also update energy production dashboards. Students gain insight into how AI supports clean energy operations.
- **Mine Operations Dispatch Assistant:** Create an agent that helps coordinate mining operations by tracking equipment availability, worker schedules, and production targets. It could automatically adjust plans when a machine breaks down or when conditions change. Students see how automation supports complex, real-time operations.
- **Inventory and Supply Replenishment Assistant:** Build an agent that tracks material usage, predicts when supplies will run out, and drafts restock requests. It can update inventory systems and notify procurement teams. Students learn how automation supports supply chain management.